

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.Tech AI/DS	Date:	

Code of Conduct

I Nirmal Kasi 192424248 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nirmal

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Q1. Constituency Parsing vs Dependency Parsing

Constituency Parsing

- Represents a sentence as **hierarchical phrases** such as NP, VP and PP.
- Shows how words combine to form larger grammatical units.
- Example:
 $S \rightarrow NP + VP$
- Useful for identifying phrase-level structures and nested relationships.

Dependency Parsing

- Represents syntax through **word-to-word dependencies**.
- Each word is connected to its head using dependency relations.
- Example:
chasing → subject → dog
chasing → object → cat
- Directly captures relationships such as subject, object and modifier. ([MIT Press Direct](#))

Comparison

Feature	Constituency	Dependency
Basic unit	Phrase	Word
Representation	Hierarchical tree	Dependency tree
Focus	Phrase structure	Word relationships
SVO extraction	Requires extra processing	Direct
Long-distance relationships	Can represent	Easier to represent directly
Best for	Phrase-level analysis	Relationship extraction

Conclusion: Dependency parsing is more suitable for extracting meaningful relationships from conversational text because it directly represents relationships between words, making subject-verb-object and other semantic relationships easier to extract. ([Baeldung on Kotlin](#))

Q2. Top-Down Parsing vs Earley Parsing

For incomplete queries such as:

- “**best hotels near...**”
- “**book a flight from Chennai to...**”

Top-Down Parsing

- Starts from the start symbol and predicts grammar rules.
- Can require considerable backtracking when several interpretations are possible.
- Incomplete input can make prediction difficult.
- Less suitable for interactive systems requiring quick responses.

Earley Parsing

- Uses chart-based parsing.
- Maintains partial parse states.
- Can handle ambiguous and incomplete input.
- Avoids repeatedly processing the same partial structures.
- Works with general context-free grammars. ([Computational Linguistics](#))

Comparison

Feature	Top-Down	Earley
Incomplete input	Poor	Good
Ambiguity	Backtracking	Maintains alternatives
Partial parsing	Limited	Good
Efficiency	Can become inefficient	More efficient for general CFG
Interactive NLP	Less suitable	More suitable

Conclusion: Earley Parsing is more appropriate for interactive NLP applications because it can maintain partial interpretations and handle incomplete or ambiguous queries effectively.

Q3. CFG vs PCFG vs Neural Constituency Parser

CFG

- Uses fixed grammar rules.
- Generates possible syntactic parse trees.
- Highly interpretable.
- Does not naturally determine which ambiguous parse is most likely.

PCFG

- Extends CFG by assigning probabilities to grammar rules.
- Each parse receives a probability.
- The highest-probability parse can be selected.
- Better for resolving structural ambiguity than ordinary CFG. ([CMU Machine Learning Blog](#))

Neural Constituency Parser

- Uses neural models to learn syntactic patterns from data.
- Can capture complex contextual information.
- Generally provides better practical accuracy.
- Less interpretable than rule-based CFG approaches.

Comparison

Feature	CFG	PCFG	Neural Parser
Interpretability	High	High	Lower
Ambiguity handling	Weak	Good	Very good
Uses probabilities	No	Yes	Yes/learned scores
Context understanding	Limited	Limited	Strong
Practical accuracy	Moderate	Better	Usually highest
Data requirement	Low	Training data useful	High

Conclusion: PCFG provides the best balance between interpretability and practical accuracy because it retains grammar-based structure while using probabilities to select among ambiguous parses.

Q4. Feature Structures vs Traditional CFG

Traditional CFG

CFG represents grammatical structures using rules such as:

However, ordinary CFG does not conveniently encode features such as:

- Number
- Gender
- Person
- Tense

This can require a very large number of grammar rules. ([Computer Science & Technology Dept.](#))

Feature Structures

Feature structures attach grammatical information to linguistic categories.

Example:

NP [NUMBER = singular, PERSON = 3]

VP [NUMBER = singular]

The features can be **unified** so that incompatible grammatical combinations are rejected.

Comparison

Feature	CFG	Feature Structures
Basic grammar	Yes	Yes
Number agreement	Limited	Strong
Gender	Difficult	Easy
Person	Difficult	Easy
Tense	Limited	Easy
Grammar size	Can become large	More compact
Constraint enforcement	Weak	Strong

Conclusion: Feature structures are more useful for multilingual NLP because they can explicitly represent and enforce grammatical constraints such as **number, gender, tense and person** without requiring huge numbers of separate grammar rules.

Q5. Subcategorization Frames

A **subcategorization frame** specifies the arguments that a verb requires to form a valid sentence.

Examples

1. “Give the book to John.”

give → Subject + Object + Recipient

- Subject = implied giver
- Object = book
- Recipient = John

2. “Sleep.”

sleep → Subject

The verb does not require an object.

3. “Put the file on the desk.”

put → Subject + Object + Location

- Object = file
- Location = desk

Importance

Subcategorization helps the NLP system determine:

- Which arguments a verb requires.
- Which phrases are valid with the verb.
- The semantic role of each argument.
- Whether a sentence is syntactically complete.
- How to construct the correct semantic representation.

Feature-based linguistic representations can encode information such as verb subcategorization and argument requirements.

Conclusion: Subcategorization frames are important because they connect **verb meaning with syntactic structure**, helping a voice assistant correctly interpret who performs an action, what is affected, and where or to whom the action applies.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1	4	3	4	4	4	19	
2	3	4	4	4	4	19	
3	4	4	4	4	3	19	
4	4	3	4	4	4	19	
5	4	4	4	3	4	19	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____